HW3

1) Let us assume that the DFA must be M, with states $\{X, Y\}$, where X is the starting state and Y is the accepting state and transitions:

→ a toggles: $X \xrightarrow{a} Y$, $Y \xrightarrow{a} X$
→ b loops: $X \xrightarrow{b} X$, $Y \xrightarrow{b} Y$

Let the second DFA be M_2 with states $\{1, 2, 3, 4\}$, start 1, accept 4, and has transitions:

From 1: $a \to 2$, $b \to 1$
From 2: $a \to 2$, $b \to 3$
From 3: $a \to 4$, $b \to 1$
From 4: $a \to 4$, $b \to 4$

Now let us construct the cross product for all the parts. Over here, the DFA M has:

→ States: $\{X, Y\} \times \{1, 2, 3, 4\}$
Start: $\{X, 1\}$
Alphabet: $\{a, b\}$

Transition: $\delta((p,i),\sigma) = (\delta_1(p,\sigma), \delta_2(i,\sigma))$

Since, we are looking for rows over here, we can follow the following product transitions:

State	On a →	On b →
(X,1)	(Y,2)	(X,1)
(X,2)	(Y,2)	(X,3)
(X,3)	(Y,4)	(X,1)
(X,4)	(Y,4)	(X,4)
(Y,1)	(X,2)	(Y,1)
(Y,2)	(X,2)	(Y,3)
(Y,3)	(X,4)	(Y,1)
(Y,4)	(X,4)	(Y,4)

a) Now, in our product DFA, a state (p,i) is accepting iff $p \in \{Y\}$ or $i \in \{4\}$, hence our accepting states are:

$F_U = \{(Y,1), (Y,2), (Y,3), (Y,4), (X,4)\}$.

b) ~~letus~~ It only accepts iff $p \in \{Y\}$ and $i \in \{4\}$, so $F_\cap = \{(Y,4)\}$ and transitions are unchanged

c) It only accepts iff exactly one of these states is true: $p \in \{Y\}$ or $i \in \{4\}$. So, we get:

$$F_\oplus = \{(Y,1),(Y,2),(Y,3),(X,4)\}$$

(We haven't included $(Y,4)$ because its not included in both).

2) Claim: In general, for an NFA $N = (Q, \Sigma, \delta, q_0, F)$ if we flip accepting and non accepting states to get $N' = (Q, \Sigma, \delta, q_0, Q/F)$, then $L(N')$ is not necessarily the complement of $L(N)$.

~~letus~~ Now, let us assume that $\Sigma = \{a\}$ and let us define an NFA N such that:

→ $Q = \{q_0, q_1\}$

→ start state: q_0

- accept states $F = \{q_1\}$
- transitions:

1) $\delta(q_0, a) = \{q_0, q_1\}$
2) $\delta(q_1, a) = \{q_1\}$

Now, let us try to figure out what language does N accept over here:

→ For the string ε: it only end at q_0, which does not accept, so $\varepsilon \notin L(N)$.

→ For any string a^n with $n \geq 1$: on the first a, N can go to q_1, and then stay in q_1 forever, so there exists an accepting path.

Thus; $L(N) = \{a^n : n \geq 1\} = a^+$

Now, let us compare it with the true complement. So, over here the complement of $L(N) = a^+$ over $\Sigma = \{a\}$ is $\overline{L(N)} = \{\varepsilon\}$, but we found that $L(N') = a^* \neq \{\varepsilon\}$.

Therefore, we can say that L(N') is not always a complement of L(N).

Conclusion: We can see that flipping accept / non accept states works for NFA DFAs (complete deterministic machines), but fails for NFAs because the acceptance says that "there exists an accepting path" and flipping states doesn't turn that into "no accepting paths"

5) Start state: 1
Accept state: 5
ε-moves: $1 \xrightarrow{\varepsilon} 2$, $1 \xrightarrow{\varepsilon} 4$, $3 \xrightarrow{\varepsilon} 1$
Other moves:

- $2 \xrightarrow{a} 3$, $3 \xrightarrow{a} 2$
- Loops: $2 \xrightarrow{b} 2$, $3 \xrightarrow{b} 3$, $4 \xrightarrow{b} 4$, $5 \xrightarrow{a} 5$
- $4 \xrightarrow{a} 5$, $5 \xrightarrow{b} 4$

Now, we can say that the DF start state is 124, since ε-closure $(\{1\}) = \{1, 2, 4\}$

Now, using the subset construction, our reachable DFA states are: {124, 24, 34, 25, 345}

Over here, we know that a DFA is reachable iff it contains 5, hence we can say that our accepting DFA states are {25, 345}.

Hence our final DFA table can be shown in the following way:

DFA state	On a →	on b →
124 (start)	345	24
24	345	24
34	25	34
25 (accept)	345	24
345 (accept)	25	34